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AI-empowered Next Generation Computer-Aided Engineering

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Jiachen Guo

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ABSTRACT

AI-empowered Next Generation Computer-Aided Engineering

Jiachen Guo

Traditional Computer-Aided Engineering (CAE) faces a critical challenge: the prohibitive computational cost of high-fidelity simulations. As engineering designs in additive manufacturing and integrated circuits increase in complexity, standard methods struggle to provide the agility required for real-time digital twins. This work addresses these challenges by establishing a new paradigm of AI-empowered computational mechanics, integrating tensor order reduction methods with agentic reasoning to deliver massive gains in both performance and accessibility.

The core of this framework is the Tensor-decomposition-based A Priori Surrogate (TAPS), a data-free, physics-based method designed to overcome the curse of dimensionality in parametric design. By integrating AI-enhanced finite element interpolants (C-HiDeNN) with low-rank tensor approximations, TAPS solves the generalized Space-Parameter-Time weak form directly. This approach preserves the rigor of classical finite element methods while enabling the simulation of zetta-scale degrees of freedom on a single GPU with minimal memory requirements.

To bridge the substantial implementation gap of such sophisticated solvers, we introduce an LLM-driven agentic workflow for computational mechanics. We demonstrate that Large Language Models, guided by Chain-of-Thought (CoT) prompting, can autonomously derive complex variational weak forms and generate high-performance GPU code. We validate this approach through the automated implementation of the MultiLevel Variational MultiScale (ML-VMS) method. By using the LLM agent to handle the intrusive programming and cross-scale coupling required for ML-VMS, we demonstrate that complex, high-order multiscale solvers can be deployed rapidly without manual derivation, effectively democratizing state-of-the-art CAE tools.

Finally, we resolve the geometric pre-processing bottleneck through the Immersed Tensor Decomposition (ITD) framework. By replacing traditional body-fitted meshing with an immersed voxelated representation and optimized Signed Distance Function (SDF) parameterization, ITD enables the automated analysis of complex CAD geometries. Combined with the immersed tensor decomposition (ITD) solver, this approach enables seamless simulation of massive domains with more than 300 million degrees of freedom. Together, these contributions provide a scalable, intelligent ecosystem for next-generation CAE.

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Preface

This dissertation presents research on AI-empowered numerical methods conducted during the author’s doctoral studies. The chapters presented here are adapted from the following articles:

[1] Guo, J., Domel, G., Park, C., Zhang, H., Gumus, O.C., Lu, Y., Wagner, G.J., Qian, D., Cao, J., Hughes, T.J. and Liu, W.K., 2025. Tensor-decomposition-based A Priori Surrogate (TAPS) modeling for ultra large-scale simulations. *Computer Methods in Applied Mechanics and Engineering*, 444, p.118101.

[2] Guo, J., Xie, X., Park, C., Zhang, H., Politis, M., Domel, G., and Liu, W.K., 2025. Interpolating Neural Network-Tensor Decomposition (INN-TD): a scalable and interpretable approach for large-scale physics-based problems. *Proceedings of the 42nd International Conference on Machine Learning*, PMLR, 21138–21162.

[3] Guo, J., Park, C., Qian, D., Hughes, T.J. and Liu, W.K., 2026. Large language model-empowered next-generation computer-aided engineering. *Computer Methods in Applied Mechanics and Engineering*, 450, p.118591.

[4] Guo, J., Park, C., Xie, X., Sang, Z., Wagner, G.J. and Liu, W.K., 2024. Convolutional Hierarchical Deep Learning Neural Networks-Tensor Decomposition (C-HiDeNN-TD): a scalable surrogate modeling approach for large-scale physical systems. *NeurIPS 2024 workshop on Data-driven and Differentiable Simulations, Surrogates, and Solvers*

[5] Zhang, L., Guo, J., Tang, S., Hughes, T.J. and Liu, W.K., 2025. MultiLevel Variational MultiScale (ML-VMS) framework for large-scale simulation. *Computer Methods in Applied Mechanics and Engineering*, 453, 118807.

[6] Guo, J., and Liu, W.K., 2026. Immersed Tensor Decomposition for large-scale analysis of irregular geometries, in preparation.

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PREVIEW

CHAPTER 1

Introduction

Precision is a fundamental aspect of scientific and engineering applications, particularly in advanced industries such as semiconductor manufacturing and additive manufacturing. The capability to perform accurate computational simulations is essential for advancing these fields, as it enables the optimization of design and manufacturing processes through virtual prototyping. This approach reduces the reliance on expensive physical testing and provides critical insights where physical prototypes are impractical. However, traditional computational methods, such as the Finite Element Method (FEM), incur prohibitive computational costs when predicting responses across multiple length and time scales [6]. Achieving high precision typically requires increasing mesh resolution, which increases computational costs and Random-Access Memory (RAM) requirements that far exceed the capabilities of standard workstations, often necessitating massive parallelization on supercomputers. In industries like additive manufacturing, the vast design space further exacerbates these limitations, as exploring the effects of numerous design parameters requires numerous expensive simulations.

To address the growing challenges in predictive scientific modeling, data-driven surrogates—especially those based on Artificial Intelligence (AI)—have emerged as an alternative to conventional numerical models. These models offer significantly reduced forward prediction times and can serve as reduced representations of real physics [14, 15]. However, standard AI models often fail to meet the high accuracy requirements of

predictive scientific tasks [16]. Furthermore, the widely adopted offline-online training scheme creates a new bottleneck: the need to generate massive datasets for training. For high-dimensional design spaces governed by parameterized partial differential equations (PDEs), generating this training data via repetitive expensive simulations is often infeasible. To circumvent these costs, this thesis explores a paradigm shift toward data-free, or “A Priori,” reduced-order models. Unlike data-driven approaches, a priori methods derive surrogate models directly from the governing equations without the need for pre-generated data.

Recent developments in Hierarchical Deep-learning Neural Networks (HiDeNN) and Convolutional HiDeNN (C-HiDeNN) have successfully incorporated FEM principles into neural architectures to enhance accuracy and efficiency [10, 7, 12]. However, even these advanced architectures struggle with the “curse of dimensionality” in ultra-large-scale problems. To resolve this, this research leverages Model Order Reduction (MOR) techniques, specifically Tensor Decomposition (TD). By combining C-HiDeNN with TD, this work addresses large-scale problems previously considered intractable. The proposed Tensor-decomposition-based A Priori Surrogate (TAPS) aims to enhance high-resolution capabilities while simultaneously optimizing computational efficiency with a minimal memory footprint.

While the mathematical foundation of data-free intrusive MOR is robust, its practical adoption is hindered by the complexity of implementation. Developing these solvers requires extensive mathematical derivation, given the new parametric problem statement, which poses a high barrier to entry. We are currently witnessing a paradigm shift in software development where natural language serves as the programming interface [17].

To address the implementation barrier, this thesis introduces a Large Language Model (LLM)-empowered Computer-Aided Engineering (CAE) agent. Unlike previous applications that focus on automating workflows using existing software, this work presents an agent capable of automating the development of the solvers themselves. By leveraging the reasoning capabilities of LLMs, we automate the mathematical derivation and implementation of TAPS solvers, significantly reducing human effort and democratizing access to advanced solver technology.

Beyond the solver formulation, modern engineering challenges often involve complex multiscale physics and irregular geometries. For example, processes like Laser Powder Bed Fusion (LPBF) require resolving microscale melt pool physics alongside macroscale heat transfer [18]. Standard numerical tools often fail to efficiently resolve these cross-scale physics. To address this, we introduce the Multilevel Variational MultiScale (ML-VMS) method, utilizing C-HiDeNN interpolations to seamlessly bridge scales without the prohibitive cost of uniform fine meshing. Furthermore, industrial applications often involve “dirty” geometries—such as CT scans of biological tissues or voxel-native AM parts—that are difficult to mesh using body-fitted approaches. We propose the Immersed Tensor Decomposition (ITD) framework, which utilizes immersed voxelated geometric representations and Signed Distance Functions (SDFs) to bypass the meshing bottleneck entirely while maintaining high fidelity.

This thesis is structured to address these challenges systematically. We first introduce the TAPS formulation in Chapter 2 and examine its numerical convergence for space-time and space-parameter-time problems. Chapter 3 presents the LLM-Empowered CAE Agent, demonstrating how Generative AI can automate the complex development and

implementation of TAPS solvers. Chapter 4 details the ML-VMS Framework, applying C-HiDeNN and tensor decomposition to solve multiscale problems. Chapter 5 proposes the Immersed Tensor Decomposition (ITD) framework, offering a robust solution for simulating complex, voxelated geometries without the need for body-fitted meshing. Finally, Chapter 6 provides a summary of the proposed method and outlines future directions.

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